

Jiechao Gao

CONTACT INFORMATION	 +1 929-213-7363	Y2E2
	 jiechao@stanford.edu  linkedin.com/in/jiechaogao  Scholar  Homepage	Center for Sustainable Development & Global Competitiveness Stanford University Stanford Profile Stanford, CA
STATUS	On the 2026–27 academic job market. U.S. Permanent Resident, no visa sponsorship required.	
RESEARCH INTERESTS	I build AI systems that stay reliable under real-world deployment constraints: limited resources, distributed private data, and high-stakes outputs. • Trustworthy Generation & Agents: verifiable multi-framework code generation (UI2Code), factuality & hallucination evaluation, LLM-as-judge, neuro-symbolic interpretability (ICML’26 ×2, AAAI’26 ×2, WWW’26, ACL’26, Findings of ACL’26, ICLR’26) • Industrial-Scale Multimodal Alignment & Recommendation: ID-multimodal representation alignment and distribution-aware modeling in billion-user platforms (NeurIPS’25, SIGIR’26, KDD’25 ×2, ICML’25, ACM MM’25 ×2) • Privacy-Preserving & Efficient Learning Systems: federated learning, on-device personalization, differential privacy, LLM efficiency & economic viability (NPJ AI’26, ICCPS’25, SenSys, IEEE TSC’20, IPDPS’20, EMNLP’25) • AI for High-Stakes Domains & Open Source: healthcare, smart buildings & sustainable energy, finance (FinRL ecosystem) (AAAI’26, EMNLP’25 Best Paper Nom., CHIL’23, BuildSys’24, NeurIPS’22 D&B)	
PROFESSIONAL EXPERIENCE	Stanford University Postdoc Research Associate Jul. 2025 - now	• Research on trustworthy generative models and resource-efficient LLM systems, with applications in engineering and sustainability. • Advisors: Prof. Jie Wang and Prof. Michael Lepech
	Columbia University Associate Research Scientist Aug. 2024 - June 2025	• Federated learning, reinforcement learning, and trustworthy LLM agents for financial applications; core contributor and contest organizer of the FinRL open-source ecosystem.
INDUSTRY EXPERIENCE	Frees Fund (VC), Beijing AI Investment Intern May 2018 - Aug. 2018	
EDUCATION	Ph.D. in Computer Science <i>University of Virginia</i> Department of Computer Science Aug. 2018 - Aug. 2024 Charlottesville, VA	
	Master of Science in Electrical Engineering <i>Columbia University</i> Department of Electrical Engineering Sep. 2016 - Feb. 2018 New York, NY	
	B.S. in Applied Physics & B.Eng. in Financial Engineering <i>Jilin University</i> Department of Physics & Department of Economics Sep. 2012 - Jul. 2016 Changchun, Jilin	

AWARDS

Google HE Faculty AI Fellowship	2026
EMNLP Industry Track Best Paper Nomination	2025
World's Top 2% Scientists (Stanford/Elsevier)	2024 & 2025
ACM BuildSys Best Paper Candidate	2024
ICPP NSF Travel Award	2023
UVA Link Lab Flash Talk Awardee	Feb. 2022
Excellent Graduation Thesis (5/242), JLU Physics Department	Jun. 2016
MCM Outstanding Winner Award	Apr. 2015
National Scholarship (Top 1%)	2013 & 2014
Manulife Financial Center Outstanding Intern Award	2014

PUBLICATIONS



4,616 Citations · H-index 29 · i10-index 54 (Google Scholar, as of Aug. 2026)

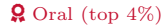
106 peer-reviewed publications (62 conference · 30 journal · 11 workshop/poster/demo · 3 book chapters) + 8 preprints + 11 main-track submissions under review, with **66 as first / co-first / corresponding author**. Venues include **ICML ×3, NeurIPS ×2 (main and D&B), ICLR, AAAI ×2, KDD ×3, WWW, SIGIR, ACM MM ×3, CVPR, the ACL family ×12 (ACL / EMNLP / EACL, incl. Industry & Findings), NPJ Artificial Intelligence, ICCPS, CHIL, IPDPS, and IEEE TSC**; with 1 Best Paper Nomination (EMNLP Industry '25), 1 Best Paper Candidate (BuildSys '24), and 2 Orals (ACM MM '25 top 4%; CHIL '23 13.3%).

* equal contribution † corresponding author

Peer-reviewed Conference Papers (with proceedings)

ICML '26	62. Zixiong Yang*, Linxiao Li*, Jiaye Lin, Binrui Wu, Xiaoyu Kang, Jiechao Gao [†] Deterministic Component Mining for Multi-framework UI2Code Generation <i>International Conference on Machine Learning</i> , 2026
ICML '26	61. Jie Wu*, Haoran Ma*, Shisong Tang, Yulin Xu, Xiaoyu Kang, Jiechao Gao [†] MulFCoder: Framework-conditioned Multi-agent for MLLM-based Multi-framework Front-end Code Generation <i>International Conference on Machine Learning</i> , 2026
ICLR '26	60. Siyuan Chen*, Hang Ding*, Kangxiao Yu, Jiechao Gao [†] GRO-RAG: Gradient-aware Re-rank Optimization for Multi-source Retrieval-Augmented Generation <i>International Conference on Learning Representations</i> , 2026
AAAI '26	59. Jiechao Gao [†] , Chang Liu, Yuangang Li S2D-ALIGN: Shallow-to-Deep Auxiliary Learning for Anatomically-Grounded Radiology Report Generation <i>AAAI Conference on Artificial Intelligence</i> , 2026
AAAI '26	58. Yuangang Li*, Yiqing Shen*, Yi Nian, Jiechao Gao , Ziyi Wang, Chenxiao Yu, Shawn Li, Jie Wang, Xiyang Hu [†] , Yue Zhao [†] Mitigating hallucinations in large language models via causal reasoning <i>AAAI Conference on Artificial Intelligence</i> , 2026
KDD '26 Blue Sky	57. Yixiang Yao, Yuhang Yao, Xinyi Fan, Jiechao Gao , Jie Wang, Minjia Zhang, Srivatsan Ravi, Carlee Joe-Wong Trustworthy Agent Network: Trust in Agent Networks Must Be Baked In, Not Bolted On <i>ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Blue Sky Track</i> , 2026
WWW '26	56. Sahil Tripathi, Gautam Siddharth Kashyap, Mehwish Nasim, Jian Yang, Jiechao Gao [†] , Usman Naseem They Said Memes Were Harmless-We Found the Ones That Hurt: Decoding Jokes, Symbols, and Cultural References <i>The ACM Web Conference</i> , 2026
SIGIR '26	55. Qi Sun, Yulin Xu, Zelin Wang, Xiaoyu Kang, Keyan Jin, Jiechao Gao [†] Distribution-aware Re-representations for Multi-Scenario Recommendations <i>International ACM SIGIR Conference on Research and Development in Information Retrieval</i> , 2026

- ACL '26 Ind.** 54. Abdullah Mohammad, Sushant Kumar Ray, Pushkar Arora, Rafiq Ali, Ebad Shabbir, Gautam Siddharth Kashyap, **Jiechao Gao**[†], Usman Naseem[†]
Are Large Language Models Economically Viable for Industry Deployment?
Annual Meeting of the Association for Computational Linguistics, Industry Track, 2026
- ACL '26 Ind.** 53. Soumedhik Bharati*, Ebad Shabbir*, **Jiechao Gao**[†]
FROST: Factual Reasoning via Optimized Stochastic Trajectories in Large Language Models during Inference
Annual Meeting of the Association for Computational Linguistics, Industry Track, 2026
- ACL Findings '26** 52. **Jiechao Gao**[†], Rohan Kumar Yadav, Yuangang Li, Yuandong Pan, Jie Wang, Ying Liu, Michael Lepech
LLM-Guided Semantic Bootstrapping for Interpretable Text Classification with Tsetlin Machines
Findings of the Association for Computational Linguistics: ACL 2026
- EMNLP '26 Ind.** 51. Muran Yu, **Jiechao Gao**, Yuandong Pan, Barney Haoyun Miao, Andrew Carter Lesh, Kincho Law, Jie Wang, Michael Lepech
EGT-KG: Evidence-Grounded Typed KG Retrieval for Practical Scientific QA with Small Language Models
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2026
- EACL '26 Ind.** 50. Sushant Kumar Ray, Gautam Siddharth Kashyap, Sahil Tripathi, Nipun Joshi, Vijay Govindarajan, Rafiq Ali, **Jiechao Gao**[†], Usman Naseem
Do Clinical Question Answering Systems Really Need Specialised Medical Fine Tuning?
Conference of the European Chapter of the Association for Computational Linguistics, Industry Track, 2026
- EACL '26 Find.** 49. Gautam Siddharth Kashyap, Harsh Joshi, Niharika Jain, Ebad Shabbir, **Jiechao Gao**[†], Nipun Joshi, Usman Naseem[†]
Revealing the Truth with ConLLM for Detecting Multi-Modal Deepfakes
Conference of the European Chapter of the Association for Computational Linguistics, 2026
- ICASSP '26** 48. Ebad Shabbir*, Pushkar Arora*, Rakshita Saksainaa, Tiange Xie, **Jiechao Gao**[†]
AVATAR: Audio-Visual Adaptive Fusion via Trained Agent Reinforcement for Multimodal Deepfake Detection
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026
- ICASSP '26** 47. Yuheng Li*, Taoyu Wu*, **Jiechao Gao**[†], Wei Zhu[†], Wenwen Ouyang
DemoReranker: Enhancing the In-Context Learning Capability of Multi-Modal Large Models via Demo Ranking
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026
- ICASSP '26** 46. Xinyuan Huang, **Jiechao Gao**[†], Tiange Xie
Differentiable Meta-Optimization for Federated Neural Architecture Search (MO-FedNAS)
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026
- ICASSP '26** 45. Taoyu Wu, Yuheng Li, **Jiechao Gao**[†], Wei Zhu[†], Ming Tan, Hui Yi Leong
EPO: Enhanced Preference Optimization with Multi-Response Data for LLMs via Stochastic Softmax
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026
- ICML '25** 44. Shisong Tang*, Chao Cui*, Fan Li, **Jiechao Gao**, Hechang Chen
Calibrating Video Watch-time Predictions with Credible Prototype Alignment
International Conference on Machine Learning, 2025
- NeurIPS '25** 43. Yidong Wu*, Siyuan Chen*, Binrui Wu, Fan Li, **Jiechao Gao**[†]
Adaptive Gradient Masking for Balancing ID and MLLM-based Representations in Recommendation
Conference on Neural Information Processing Systems, 2025
- KDD '25** 42. Yulin Xu*, Chao Cui*, Shisong Tang*, Fan Li, Bing Han, Huafeng Cao, **Jiechao Gao**[†], Hechang Chen
Leveraging Label Distributions as Anchors to Enhance Video Recommendation
ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025
- KDD '25** 41. Binrui Wu*, Shisong Tang*, Fan Li, Bing Han, Chang Meng, Jingyu Xiao, **Jiechao Gao**[†]
Aligning and Balancing ID and Multimodal Representations for Recommendation
ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025
- ACL Findings '25** 40. Xunzhu Tang*, **Jiechao Gao***, Jin Xu*, Tiezhu Sun, Yewei Song, Saad Ezzini, Wendkūuni C. Ouédraogo, Jacques Klein, Tegawendé F. Bissyandé
SynFix: Dependency-Aware Program Repair via RelationGraph Analysis
Findings of the Association for Computational Linguistics: ACL 2025

- EMNLP '25 Ind.  Best Paper Nom.
39. Mohammad Anas Azeez*, Rafiq Ali*, Ebad Shabbir, Zohaib Hasan Siddiqui, Gautam Siddharth Kashyap, **Jiechao Gao**[†], Usman Naseem[†]
Truth, Trust, and Trouble: Medical AI on the Edge
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025
- EMNLP '25 Ind.
38. Zohaib Hasan Siddiqui*, **Jiechao Gao***, Ebad Shabbir*, Mohammad Anas Azeez, Rafiq Ali, Gautam Siddharth Kashyap, Usman Naseem[†]
LLMs on a Budget? Say HOLA
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025
- EMNLP '25 Ind.
37. Hui Yi Leong, Yuheng Li, Yuqing Wu, Wenwen Ouyang, Wei Zhu[†], **Jiechao Gao**[†], Wei Han
AMAS: Adaptively Determining Communication Topology for LLM-based Multi-agent System
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025
- EMNLP '25 Ind.
36. Yuheng Li*, **Jiechao Gao***, Wei Han, Wenwen Ouyang, Wei Zhu[†], Hui Yi Leong
FT-MDT: Extracting Decision Trees from Medical Texts via a Novel Low-rank Adaptation Method
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025
- IJCNLP-AAACL '25
35. Sahil Tripathi, MD Tabrez Nafis, Imran Hussain, **Jiechao Gao**[†]
The Confidence Paradox: Can LLM Know When It Is Wrong?
International Joint Conference on Natural Language Processing and AAACL, 2025
- ACM MM '25  Oral (top 4%)
34. Binrui Wu*, Haochen Sui*, Jiaye Lin*, **Jiechao Gao**[†], Ting Xu, Keyan Jin, Xuesong Zhang
Prototype-Guided Representation Projection for Multi-Domain Multi-Task Recommendation
ACM International Conference on Multimedia, 2025
- ACM MM '25
33. Songning Lai*, Ninghui Feng*, **Jiechao Gao***, Hao Wang, Haochen Sui, Xin Zou, Jiayu Yang, Wenshuo Chen, Lijie Hu, Hang Zhao, Xuming Hu, Yutao Yue
From Guesswork to Guarantee: Towards Faithful Multimedia Web Forecasting with TimeSieve
ACM International Conference on Multimedia, 2025
- CVPR '25
32. Yuzhi Huang*, Chenxin Li[†], Haitao Zhang, Zixu Lin, Yunlong Lin, Hengyu Liu, Wuyang Li, Xinyu Liu, **Jiechao Gao**, Yue Huang[†], Xinghao Ding, Yixuan Yuan
Track Any Anomalous Object: A Granular Video Anomaly Detection Pipeline
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025
- ICCPS '25
31. **Jiechao Gao***, Mingyue Tang*, Wenpeng Wang, Tushar Routh, Brad Campbell
Atlas: Ensuring Accuracy for Privacy-Preserving Federated IoT Applications
ACM/IEEE International Conference on Cyber-Physical Systems, 2025
- ICASSP '25
30. Gautam Kashyap, Zohaib Hasan Siddiqui, Mohammad Anas Azeez, Rafiq Ali, Shantanu Kumar, Navin Kamuni, **Jiechao Gao**[†]
Fooling the Forgers: A Multi-Stage Framework for Audio Deepfake Detection
IEEE International Conference on Acoustics, Speech and Signal Processing, 2025
- ICASSP '25
29. Gautam Kashyap, Harsh Joshi, Manaswi Kulahara, Rajkumar Dhakar, Atul Sajjanhar, **Jiechao Gao**[†], Sarthak Jain, Shahab Sohail
Can AI See What We Can't? Leveraging Deep Learning and Multi-Temporal Satellite Data to Revolutionize Crop Type Mapping and Yield Prediction
IEEE International Conference on Acoustics, Speech and Signal Processing, 2025
- BigData '25
28. **Jiechao Gao**, Rohan Kumar Yadav, Xinyuan Huang, Jie Wang
Enhancing Interpretability in Self-Training with Tsetlin Machines for Mitigating Noisy Pseudo-Labels
IEEE International Conference on Big Data, 2025
- BigData '25
27. Xinyuan Huang, **Jiechao Gao**[†], Jie Wang
Federated Neural Architecture Search with Model-Agnostic Meta Learning
IEEE International Conference on Big Data, 2025
- BIBM '25
26. Xinyuan Huang, **Jiechao Gao**[†]
Toward Fair and Efficient Neural Architecture Search in Heterogeneous Federated Learning
IEEE International Conference on Bioinformatics and Biomedicine, 2025
- BIBM '25
25. Xinyuan Huang, Hikaru Kurosawa, Jack Wunder, Sujit Patil, **Jiechao Gao**, Karthik Kuber, Jonathan C. Irish, Michael J. Daly
U-KAN: Hybrid Spatial-Functional Deep Learning for Tumor Depth Estimation in Fluorescence-Guided Cancer Surgery
IEEE International Conference on Bioinformatics and Biomedicine, 2025

- KSEM '25** 24. Huaibin Ye, Zuobin Ying, **Jiechao Gao**, Ximeng Liu, Jianping Cai
FedBCE: Rethinking Clustered Federated Learning for Better Clustering Efficiency
International Conference on Knowledge Science, Engineering and Management, 2025
- ASONAM '25** 23. Arpita Soni, Sahil Tripathi, Gautam Siddharth Kashyap, Manaswi Kulahara, Mohammad Anas Azeez, Zohaib Hasan Siddiqui, Nipun Joshi, **Jiechao Gao**[†]
Can We Predict Your Next Move Without Breaking Your Privacy?
IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining, 2025
- SEC '25** 22. Marshall Clyburn*, Nabeel Nasir*, Md Fazlay Rabbi Masum Billah, Victor Ariel Leal Sobral, **Jiechao Gao**, Fateme Nikseresht, Brad Campbell
fReeLoaders: An IoT Ecosystem for Real-Time Deadline-Driven Task Scheduling using Reinforcement Learning
ACM/IEEE Symposium on Edge Computing, 2025
- ACM MM '24** 21. Yiyong Xiao*, Kai Shu*, Haoyi Zhang*, Baohua Yin, Wai Seng Cheang, Haoyang Wang, **Jiechao Gao**[†]
EGGesture: Entropy-Guided Vector Quantized Variational AutoEncoder for Co-Speech Gesture Generation
ACM International Conference on Multimedia, 2024
- BuildSys '24** 20. Tushar Routh, Nurani Saoda, Fateme Nikseresht, Md Fazlay Rabbi Masum Billah, **Jiechao Gao**,
🏆 Best Paper
Cand. Viswajith Govinda Rajan, Bradford Campbell
ScreenSense: Screen Activity Detection in Real-World Environments with Indoor Light Sensors
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2024
- BigData '24** 19. Jiayu Wang, **Jiechao Gao**[†]
Federated Learning with Knowledge Distillation to Mitigate Catastrophic Forgetting and Data Heterogeneity in IoV Systems
IEEE International Conference on Big Data, 2024
- BIBM '24** 18. **Jiechao Gao**[†], Yuangang Li
FedMetaMed: Federated Meta-Learning for Personalized Medication in Distributed Healthcare Systems
IEEE International Conference on Bioinformatics and Biomedicine, 2024
- BIBM '24** 17. Kai Shu*, Yuzhuo Jia*, Ziyang Zhang, **Jiechao Gao**[†]
FODA-PG for Enhanced Medical Imaging Narrative Generation: Adaptive Differentiation of Normal and Abnormal Attributes
IEEE International Conference on Bioinformatics and Biomedicine, 2024
- SciSec '24** 16. Tiange Xie, Feng Liu, **Jiechao Gao**, Yinghui Wang
Integrating Consortium Blockchain and Attribute-Based Searchable Encryption for Automotive Threat Intelligence Sharing Model
International Conference on Science of Cyber Security, 2024
- ICDM SSTDM '24** 15. **Jiechao Gao**[†], Yuangang Li, Syeda Faiza Ahmed
Fed-LDR: Federated Local Data-infused Graph Creation with Node-centric Model Refinement
IEEE ICDM International Workshop on Spatial and Spatiotemporal Data Mining, 2024
- CHIL '23** 14. Mingyu Tang*, **Jiechao Gao***, Guimin Dong, Carl Yang, Bradford Campbell, Brendan Bowman,
🏆 Oral (13.3%) Jamie Marie Zoellner, Emaad Abdel-Rahman, Mehdi Boukhechba
SRDA: Mobile Sensing based Fluid Overload Detection for End Stage Kidney Disease Patients using Sensor Relation Dual Autoencoder
PMLR Conference on Health, Inference, and Learning, 2023
- ICPP '23** 13. **Jiechao Gao**, Wenpeng Wang, Fateme Nikseresht, Viswajith Govinda Rajan, Bradford Campbell
PFDR: Personalized Federated Deep Reinforcement Learning for Residential Energy Management
International Conference on Parallel Processing, 2023
- DCOSS '23** 12. Wenpeng Wang*, Fateme Nikseresht*, Viswajith Govinda Rajan, **Jiechao Gao**, Bradford Campbell
Enabling ubiquitous occupancy detection in smart buildings: A wifi ftm-based approach
International Conference on Distributed Computing in Smart Systems and the Internet of Things, 2023
- PRCV '23** 11. Wei Cai, **Jiechao Gao**
SSCL: semi-supervised contrastive learning for industrial anomaly detection
Chinese Conference on Pattern Recognition and Computer Vision, 2023
- NeurIPS D&B '22** 10. Xiao-Yang Liu*, Ziyi Xia*, Jingyang Rui, **Jiechao Gao**, Hongyang Yang, Ming Zhu, Christina Dan Wang[†], Zhaoran Wang, Jian Guo[†]
FinRL-Meta: Market Environments and Benchmarks for Data-Driven Financial Reinforcement Learning
Conference on Neural Information Processing Systems, Datasets and Benchmarks Track, 2022

- COMSNETS '22** 9. Priyan Malarvizhi Kumar, Bharat Rawal, **Jiechao Gao**
Blockchain-enabled privacy preserving of IoT data for sustainable smart cities using machine learning
International Conference on COMMunication Systems & NETworkS, 2022
- IPSN '21** 8. Md Fazlay Rabbi Masum Billah, Nurani Saoda, **Jiechao Gao**, Bradford Campbell
BLE can see: a reinforcement learning approach for RF-based indoor occupancy detection
ACM/IEEE International Conference on Information Processing in Sensor Networks, 2021
- BuildSys '21** 7. Tong Wu*, Murtadha Aldeer*, Tahiya Chowdhury, Amber Haynes, Fateme Nikseresht, Mahsa Pahlavikhah Varnosfaderani, **Jiechao Gao**, Arsalan Heydarian, Brad Campbell, Jorge Ortiz
The smart building privacy challenge
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2021
- BuildSys '21** 6. Wenpeng Wang, Zetian Liu, **Jiechao Gao**, Nurani Saoda, Bradford Campbell
UbiTrack: Enabling scalable & low-cost device localization with onboard wifi
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2021
- ICPP '21** 5. Haoyu Wang, Haiying Shen, **Jiechao Gao**, Kevin Zheng, Xiaoying Li
Multi-agent reinforcement learning based distributed renewable energy matching for datacenters
International Conference on Parallel Processing, 2021
- ICAIF '21** 4. Xiao-Yang Liu, Hongyang Yang, **Jiechao Gao**, Christina Wang
FinRL: Deep Reinforcement Learning Framework to Automate Trading in Quantitative Finance
ACM International Conference on AI in Finance, 2021
- IPDPS '20** 3. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Smartly handling renewable energy instability in supporting a cloud datacenter
IEEE International Parallel and Distributed Processing Symposium, 2020
- ICCCN '20** 2. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Machine learning based workload prediction in cloud computing
International Conference on Computer Communications and Networks, 2020
- BigData '19** 1. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Task failure prediction in cloud data centers using deep learning
IEEE International Conference on Big Data, 2019
- Peer-reviewed Journal Papers**
- NPJ AI '26** 30. **Jiechao Gao***, Yuangang Li*, Jie Wang, Michael Lepech, Yue Zhao, Brad Campbell
H-FedSN: Personalized Sparse Networks for Efficient and Accurate Hierarchical Federated Learning for IoT Applications
NPJ Artificial Intelligence, 2026
- IEEE TAI '26** 29. Gautam Siddharth Kashyap, Niharika Jain, Ebad Shabbir, Harsh Joshi, Usman Naseem, **Jiechao Gao[†]**
CLARITY: A Lightweight Multimodal Transformer for Harmful Content Detection
IEEE Transactions on Artificial Intelligence, 2026
- AutoCon '26** 28. Yuandong Pan, Mudan Wang, **Jiechao Gao**, Jie Wang, Michael D. Lepech, Ioannis Brilakis
LLM-enabled multi-agent framework for automated Scan-to-BIM and Scan-to-Graph reconstruction
Automation in Construction, 2026
- AI for Eng. '25** 27. Keyi Wang, Nikolaus Holzer, Ziyi Xia, Yupeng Cao, **Jiechao Gao**, Anwar Walid, Kairong Xiao, Xiao-Yang Liu Yanglet
FinRL Contests: Data-Driven Financial Reinforcement Learning Agents for Stock and Crypto Trading
Artificial Intelligence for Engineering, 2025
- ACM TMIS '24** 26. Xiao-Yang Liu, Rongyi Zhu, Daochen Zha, **Jiechao Gao**, Shan Zhong, Meikang Qiu
Differentially Private Low-Rank Adaptation of Large Language Model Using Federated Learning
ACM Transactions on Management Information Systems, 2024
- EPSR '24** 25. Aamir Nawaz, Hongtao Wang, Huiting Yang, Hammad Armghan, **Jiechao Gao[†]**
Risk-constrained probabilistic coordination in coupled transmission and distribution system
Electric Power Systems Research, 2024
- Veh. Comm. '24** 24. Mehdi Gheisari, Aminreza Karamoozian, **Jiechao Gao**, Hemn Barzan Abdalla, Shuja Ansari, Riaz Ullah Khan, Zhaoxi Fang
Accident reduction through a privacy-preserving method on top of a novel ontology for autonomous vehicles with the support of modular arithmetic
Vehicular Communications, 2024

- IEEE Access '24** 23. M. Mahad Malik, Abdullah Altamimi, Syed Ali Abbas Kazmi, Zafar Ali Khan, Muhammad Waleed Ansari, Kamran Mujahid, **Jiechao Gao**
A Full-Fledged, Multi-Agent System Representing the Architecture of Smart Cities by Balancing Energy With Optimal Electricity Forecasting, Integrating Individual Comfort, and Extracting Financial Gains
IEEE Access, 2024
- TPDS '23** 22. Haiying Shen, Haoyu Wang, **Jiechao Gao**, Rajkumar Buyya
An Instability-Resilient Renewable Energy Allocation System for a Cloud Datacenter
IEEE Transactions on Parallel and Distributed Systems, 2023
- MLJ '23** 21. Xiao-Yang Liu, Ziyi Xia, Hongyang Yang, **Jiechao Gao**, Daochen Zha, Ming Zhu, Christina Dan Wang, Zhaoran Wang, Jian Guo
Dynamic Datasets and Market Environments for Financial Reinforcement Learning
Machine Learning Journal, 2023
- ESWA '23** 20. Ahmad Almadhor, Rizwana Irfan, **Jiechao Gao**[†], Nasir Saleem, Hafiz Tayyab Rauf, Seifedine Kadry
E2E-DASR: End-to-end deep learning-based dysarthric automatic speech recognition
Expert Systems with Applications, 2023
- ASC '23** 19. Hafiz Tayyab Rauf, **Jiechao Gao**^{*}, Ahmad Almadhor, Ali Haider, Yu-Dong Zhang[†], Fadi Al-Turjman
Multi population-based chaotic differential evolution for multi-modal and multi-objective optimization problems
Applied Soft Computing, 2023
- CAAI TIT '23** 18. Nasir Saleem, **Jiechao Gao**^{*}, Rizwana Irfan, Ahmad Almadhor, Hafiz Tayyab Rauf, Yudong Zhang, Seifedine Kadry
DeepCNN: Spectro temporal feature representation for speech emotion recognition
CAAI Transactions on Intelligence Technology, 2023
- JAIHC '23** 17. Mehdi Gheisari, Amir Javadpour, **Jiechao Gao**[†], Aaqif Afzaal Abbasi, Quoc-Viet Pham, Yang Liu
PPDMIT: A lightweight architecture for privacy-preserving data aggregation in the Internet of Things
Journal of Ambient Intelligence and Humanized Computing, 2023
- JKE '23** 16. Morshadul Hasan, Thuhid Noor, **Jiechao Gao**, Muhammad Usman, Mohammad Zoynul Abedin
Rural consumers financial literacy and access to FinTech services
Journal of the Knowledge Economy, 2023
- TITS '22** 15. Gunasekaran Manogaran, **Jiechao Gao**^{*}, Tu N Nguyen
Optimizing resource and service allocations for IoT-assisted intelligent transportation systems
IEEE Transactions on Intelligent Transportation Systems, 2022
- TITS '22** 14. **Jiechao Gao**, Gunasekaran Manogaran, Tu N Nguyen, Seifedine Kadry, Ching-Hsien Hsu, Priyan Malarvizhi Kumar
A vehicle-consensus information exchange scheme for traffic management in vehicular ad-hoc networks
IEEE Transactions on Intelligent Transportation Systems, 2022
- IoT-J '22** 13. Zeeshan Siddiqui, **Jiechao Gao**[†], Muhammad Khurram Khan
An Improved Lightweight PUF-PKI Digital Certificate Authentication Scheme for the Internet of Things
IEEE Internet of Things Journal, 2022
- JBHI '22** 12. **Jiechao Gao**, Tu N Nguyen, Gunasekaran Manogaran, Ankit Chaudhary, Gai-Ge Wang
Redemptive resource sharing and allocation scheme for Internet of Things-assisted smart healthcare systems
IEEE Journal of Biomedical and Health Informatics, 2022
- TNSE '22** 11. Gunasekaran Manogaran, Tu N Nguyen, **Jiechao Gao**[†], Priyan Malarvizhi Kumar
Deep learning-based service distribution model for wireless network assisted Internet of Everything
IEEE Transactions on Network Science and Engineering, 2022
- ACM ToSN '22** 10. Guimin Dong, Mingyue Tang, Zhiyuan Wang, **Jiechao Gao**, Sikun Guo, Lihua Cai, Robert Gutierrez, Bradford Campbell, Laura E Barnes, Mehdi Boukhechba
Graph Neural Networks in IoT: A Survey
ACM Transactions on Sensor Networks, 2022
- TGCN '22** 9. Tu N Nguyen, **Jiechao Gao**[†], Gunasekaran Manogaran, R Dinesh Jackson Samuel, Mamoun Alazab
Transfer learning-aided collaborative computational method for intelligent transportation system applications
IEEE Transactions on Green Communications and Networking, 2022

- KBS '22** 8. **Jiechao Gao**^{*†}, Nasir Saleem^{*}, Muhammad Irfan Khattak, Hafiz Tayyab Rauf, Seifedine Kadry, Muhammad Shafi
DeepResGRU: Residual gated recurrent neural network-augmented Kalman filtering for speech enhancement and recognition
Knowledge-Based Systems, 2022
- IVC '22** 7. Nasir Saleem, **Jiechao Gao**^{*}, Muhammad Irfan, Elena Verdu, Javier Parra Fuente
E2E-V2SResNet: Deep residual convolutional neural networks for end-to-end video driven speech synthesis
Image and Vision Computing, 2022
- CEE '22** 6. Muhammad Irfan Khattak, Nasir Saleem, **Jiechao Gao**, Elena Verdu, Javier Parra Fuente
Regularized sparse features for noisy speech enhancement using deep neural networks
Computers and Electrical Engineering, 2022
- Agriculture '22** 5. Yuanzhi Pan, Hua Jin, **Jiechao Gao**[†], Hafiz Tayyab Rauf
Identification of Buffalo Breeds Using Self-Activated-Based Improved Convolutional Neural Networks
Agriculture, 2022
- Neurocomp. '22** 4. Adeel Ahmed, Khalid Saleem, Osman Khalid, **Jiechao Gao**[†], Umer Rashid
Trust-aware denoising autoencoder with spatial-temporal activity for cross-domain personalized recommendations
Neurocomputing, 2022
- JBHI '21** 3. Priyan Malarvizhi Kumar, Choong Seon Hong, Fatemeh Afghah, Gunasekaran Manogaran, Keping Yu, Qiaozhi Hua, **Jiechao Gao**
Clouds proportionate medical data stream analytics for internet of things-based healthcare systems
IEEE Journal of Biomedical and Health Informatics, 2021
- FGCS '21** 2. Mehdi Gheisari, Hamid Esmaeili Najafabadi, Jafar A Alzubi, **Jiechao Gao**[†], Guojun Wang, Aaqif Afzaal Abbasi, Aniello Castiglione
OBPP: An ontology-based framework for privacy-preserving in IoT-based smart city
Future Generation Computer Systems, 2021
- TSC '20** 1. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Task failure prediction in cloud data centers using deep learning
IEEE Transactions on Services Computing, 2020
- Peer-reviewed Workshop, Poster & Demo Papers**
- ICWSM W '26** 11. Daniel Koo, **Jiechao Gao**, Yuandong Pan, Jie Wang, Michael D. Lepech
Perceiving Creativity in the Age of AI: How Labels, Beliefs, and Familiarity Shape Evaluations of AI-Generated and Human-Created Art
Workshop on Social Influence in the Era of LLMs (SocialLLM) at ICWSM, 2026
- ICCPS '25 Poster** 10. Tushar Routh, **Jiechao Gao**, Bradford Campbell
Data Efficient PV based Indoor Event Detection
ACM/IEEE International Conference on Cyber-Physical Systems, Poster Track, 2025
- ACL '25 SRW** 9. Ellen Yi-Ge, **Jiechao Gao**, Wei Han, Wei Zhu[†]
DRUM: Learning Demonstration Retriever for Large Multi-modal Models
Annual Meeting of the Association for Computational Linguistics Student Research Workshop, 2025
- ACL '25 SRW** 8. Juyuan Zhang, **Jiechao Gao**, Wenwen Ouyang, Wei Zhu[†], Hui Yi Leong
Time-Llama: Adapting Large Language Models for Time Series Modeling via Dynamic Low-rank Adaptation
Annual Meeting of the Association for Computational Linguistics Student Research Workshop, 2025
- NeurIPS W '25** 7. Chunlin Feng, Lijian Huang, Keyi Wang, Yupeng Cao, Ming Zhu, **Jiechao Gao**, Xiao-Yang Liu
Standard Market Environments for Financial Reinforcement Learning
NeurIPS 2025 Workshop on Generative AI in Finance, 2025
- AAAI Bridge '23** 6. Berend Jelmer Dirk Gort^{*}, Xiao-Yang Liu^{*}, Xinghang Sun, **Jiechao Gao**, Shuaiyu Chen, Christina Dan Wang.
Deep Reinforcement Learning for Cryptocurrency Trading: Practical Approach to Address Backtest Overfitting
AAAI Bridge Program on AI for Financial Services, 2023
- SenSys '22 Poster** 5. **Jiechao Gao**^{*}, Mingyue Tang^{*}, Tianhao Wang, Bradford Campbell
Pfed-ldp: A personalized federated local differential privacy framework for iot sensing data
ACM Conference on Embedded Networked Sensor Systems, Poster Track, 2022

- IPSN '22 Poster** 4. **Jiechao Gao**, Wenpeng Wang, Bradford Campbell
Residential Energy Management System Using Personalized Federated Deep Reinforcement Learning
ACM/IEEE International Conference on Information Processing in Sensor Networks, Poster Track, 2022
- NeurIPS '21 W** 3. Xiao-Yang Liu, Jingyang Rui, **Jiechao Gao**, Liuqing Yang, Hongyang Yang, Zhaoran Wang, Christina Dan Wang, Jian Guo
FinRL-Meta: A Universe of Near-Real Market Environments for Data-Driven Deep Reinforcement Learning in Quantitative Finance
Conference on Neural Information Processing Systems, 2021
- GLOBECOM W '21** 2. M Poongodi, Mounir Hamdi, **Jiechao Gao**, Hafiz Tayyab Rauf
A Novel Security Mechanism of 6G for IMD using Authentication and Key Agreement Scheme
IEEE Global Communications Conference Workshops, 2021
- SenSys W '21** 1. **Jiechao Gao**, Wenpeng Wang, Zetian Liu, Md Fazlay Rabbi Masum Billah, Bradford Campbell
Decentralized federated learning framework for the neighborhood: a case study on residential building load forecasting
ACM Conference on Embedded Networked Sensor Systems, AIChallengeIoT Workshop, 2021

Peer-reviewed Book Chapters

- Book '22** 3. **Jiechao Gao**, Tiange Xie
DNA computing in cryptography
Advances in Computers (Elsevier), 2022
- Book '20** 2. Hongzhuo Qi, Yi Qin, **Jiechao Gao**
Cloud Computing and Big Data
Harbin Institute of Technology Press, 2020
- Book '19** 1. Jinglan Wu, Dongliang Lin, **Jiechao Gao**, Ye Xu
Introduction of Bigdata and Artificial Intelligence Applications
University of Electronic Science and Technology of China Press, 2019

Preprints

- arXiv '26** 8. Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation
arXiv preprint, 2026
- arXiv '26** 7. PrivateHub: Contrastive Diffusion Model for Private Sensor-Intensive Environment Data Generation
arXiv preprint, 2026
- arXiv '26** 6. **Jiechao Gao**, Yuandong Pan, Yuangang Li, Jie Wang, Kincho Law, Michael D. Lepech
The Orchestration Gap: Why Process Automation Stalls in Operationally Complex Industries
arXiv preprint, 2026
- arXiv '26** 5. Clusters are All You Need: Pre-Training the Tsetlin Machine with Semantic Clusters from Language Models for Interpretability
arXiv preprint, 2026
- arXiv '25** 4. Gautam Siddharth Kashyap, Mohammad Anas Azeez, Rafiq Ali, Zohaib Hasan Siddiqui, **Jiechao Gao**[†], Usman Naseem
ChildGuard: A Specialized Dataset for Combatting Child-Targeted Hate Speech
arXiv preprint, 2025
- arXiv '25** 3. Harsh Joshi, Gautam Siddharth Kashyap, Rafiq Ali, Ebad Shabbir, Niharika Jain, Sarthak Jain, **Jiechao Gao**[†], Usman Naseem
Can Argus Judge Them All? Comparing VLMs Across Domains
arXiv preprint, 2025
- Preprints.org '24** 2. Artificial Intelligence-Aided Digital Twin Design: A Systematic Review and Future Directions
Preprints.org, 2024
- arXiv '23** 1. Gautam Siddharth Kashyap, Deepkashi Mahajan, Orchid Chetia Phukan, Ankit Kumar, Alexander E. I. Brownlee, **Jiechao Gao**
From Simulations to Reality: Enhancing Multi-Robot Exploration for Urban Search and Rescue
arXiv preprint, 2023

Under Review (main-track submissions)

Anchoring LLM-based Chest X-ray Report Generation via Diffusion Language Planning

Phase-wise MLLM Tuning for Multi-framework WebUI Code Generation
Toward Executable Multi-framework Front-end Code Generation with Self-Correction
Render Structure Uncertainty for HTML Repair in MLLM-based UI-to-Code Generation
When Properties Are Not Separated: Do LLMs Fail at Judging?
Clinically Grounded Curriculum Learning for Radiology Report Generation
Fine-Grained Temporal Data Augmentation for Human Activity Recognition
GRM: Graph Reward Modeling for Multimodal Retrieval-Augmented Generation
Matched-Corruption Decoding: Tracing Support in Visual Reasoning
Show Your Evidence: Evaluating Grounded Instructional Video with Video-LLMs
PrivateHub: Contrastive Diffusion Model for Private Sensor-Intensive Environment Data Generation

GRANT &
PROPOSAL
EXPERIENCE

Contributed to funded research and proposal preparation as a graduate researcher / postdoc (not PI).

NSF: 2217071

Jul. 2022 - Sep. 2027

Collaborative Research: **PPoSS: LARGE: Co-designing Hardware, Software, and Algorithms to Enable Extreme-Scale Machine Learning Systems**

PI/co-PI(s): Aidong Zhang, Matthew Dwyer, Kevin Skadron, Bradford Campbell, Ashish Venkat

USDOT

under submission

University Transportation Centers Program - Tier 1 Center
CIFE

STUDENT
MENTORSHIP

20+ students have worked with me on research projects, many of whom moved on to PhD programs. Venues in parentheses are papers they first-authored or co-authored during our collaboration.

- Muran Yu: Stanford PhD (EMNLP'26 Ind.)
- Yulin Xu: UCI CS MS → USC CS PhD (KDD'25, ICML'26, SIGIR'26)
- Yuangang Li: USC CS MS → UCI CS PhD (AAAI'26, Findings of ACL'26, NPJ AI'26, BIBM'24, ICDM'24 W)
- Tiange Xie: Columbia U MS → IIE@CAS PhD (SciSec'24, ICASSP'26 ×2, book chapter)
- Mingyue Tang: USC CS MS → UVA SE PhD → UIUC CS PhD (ICCPs'25, SenSys'22 Poster)
- Gautam Siddharth Kashyap: Jamia Hamdard → Macquarie University CS PhD (EACL'26, ICASSP'26 ×2, ACL'26, WWW'26, EMNLP'25)
- Berend Gort: Columbia U MLE → Universitat Politècnica de Catalunya CS PhD
- Yuanbo Li: Zhengzhou University CS PhD
- Kai Shu: USC CS MS (BIBM'24, ACM MM'24)
- Xinyuan Huang: U of Toronto CS MS → U of Toronto CS PhD (ICASSP'26, BigData'25, BIBM'25)
- Binrui Wu: Fudan University (KDD'25, ACM MM'25 Oral, NeurIPS'25, ICML'26)
- Siyuan Chen: University of Bristol (ICLR'26, NeurIPS'25)
- Haoran Ma: Harvard University (ICML'26)
- Linxiao Li: University of Sydney (ICML'26)
- Haochen Sui: University of Michigan (ACM MM'25 ×2)
- Qi Sun: University of Melbourne (SIGIR'26)
- Yidong Wu: Imperial College London (NeurIPS'25)
- Ebad Shabbir: DSEU-Okhla → MBZUAI (ACL'26 ×2, EACL'26, EMNLP'25 ×2, ICASSP'26)
- Rafiq Ali: DSEU-Okhla → MBZUAI (ACL'26, EACL'26, EMNLP'25 ×2, ICASSP'26)
- Harsh Joshi: Bharati Vidyapeeth's College of Engineering → UC San Diego MS (EACL'26, ICASSP'26)

TALKS AND LECTURES	Stanford University	<i>Private and Accurate IoT Applications</i>	May. 2025
	Rutgers Efficient AI Seminar	<i>Federated Learning Enabled IoT Systems</i>	May. 2024
	Commonwealth Cyber Initiative Symposium	<i>PFed-LDP: A Personalized Federated Local Differential Privacy Framework for IoT Sensing Data</i>	Apr. 2023
	Beijing Institute of Technology	<i>Towards Accurate and Private Cloud-IoT Applications</i>	Feb. 2023
	Jilin University	<i>Towards Accurate and Private Cloud-IoT Applications</i>	Feb. 2023
	Beijing University of Technology	<i>Towards Accurate and Private Cloud-IoT Applications</i>	Feb. 2023
	UVA ML Seminar	<i>Federated Learning with Privacy</i>	Aug. 2022
	UVA Link Lab	<i>Towards Accurate and Private IoT Applications</i>	Feb. 2022
	Frees Fund	<i>How Neural Networks and Deep Learning Work in the Financial Industry</i>	Aug. 2018

TEACHING EXPERIENCE	Stanford University	Stanford, CA
	Guest Lecture	Winter, 2026
	<i>Stanford University CEE 221</i>	
	University of Virginia	Charlottesville, VA
	Teaching Assistant	Fall 2020
	<i>Computer Networks</i> (Prof. Brad Campbell)	
	Teaching Assistant (helping on course topics)	Spring 2020
	<i>Advanced Computer Networks</i> (Prof. Yixin Sun)	
	Teaching Assistant	Fall 2019
	<i>Principles of Modeling for Cyber-Physical Systems</i> (Prof. Madhur Behl)	
	Columbia University	New York City, NY
	Teaching Assistant	Fall 2017
	<i>Reinforcement Learning</i> (Prof. Chong Li)	

SERVICES	Conference/Workshop/Competition Organizing Committee	
	• ASONAM Multidisciplinary Track Co-Chair	2026
	• FinRL Contest ICAIF	2023, 2024, 2025
	• UbiComp/ISWC Workshop on Design with Autistic Children	2024
	• SecureComm Workshop on S/P-IoT	2022

Area Chair and/or (Senior) Program Committee/Reviewer for Conferences and Workshops

- IJCAI 2025 (SPC)
- AAAI-AIA 2025 (SPC)
- IJCNN 2025, 2026 (AC)
- EACL Industry Track 2026 (AC)
- IEEE IDS 2025, 2026 (Track Chair)
- NeurIPS 2022, 2023, 2024, 2025, 2026
- EMNLP Industry 2026
- SIGIR 2026
- BMVC 2026
- ICML 2025, 2026
- ACL ARR 2026
- KDD 2023, 2024, 2025, 2026
- AAAI 2024, 2025, 2026
- SDM 2024
- ML4H 2023, 2024
- UbiComp/IMWUT 2025
- ICLR 2025, 2026
- MLHC 2025
- ICASSP 2025, 2026 (AC)
- CVPR 2025, 2026
- ICCV 2025
- ICCPS 2025
- ACM MM 2025, 2026
- DSNE 2023–2026
- SecureComm 2022
- 2nd QTNML@NeurIPS 2022
- DCAA@AAAI 2023
- MILET@SIGKDD 2023
- AI4TS@IJCAI 2023
- Sensys Workshop on IoT Datasets 2024
- AI4TS@SDM 2024

Journal Reviewer

- Social Network Analysis and Mining (Associate Editor)

- IEEE Transactions on Systems, Man and Cybernetics: Systems
- Journal of Machine Learning Research (JMLR)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Transactions on Intelligent Transportation Systems (TITS)
- IEEE Journal of Biomedical and Health Informatics (JBHI)
- IEEE Transactions on Cloud Computing (TCC)
- ACM Transactions on Sensor Networks (ToSN)
- Journal of Industrial Information Integration
- ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)
- Financial Innovation
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)
- Expert Systems with Applications (ESWA)
- Cluster Computing
- IEEE Sensors Journal
- IEEE Access
- Soft Computing
- Journal of Ambient Intelligence and Humanized Computing
- Multimedia Tools and Applications

REFERENCES

Prof. Bradford Campbell

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 Department of Computer Science and Electrical Engineering
 University of Virginia
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Prof. Michael Lepech

C. L. Peck, Class of 1906 Professor (Postdoc Advisor)
 Department of Civil and Environmental Engineering
 Stanford University
 Reference email: mlepech@stanford.edu

Prof. Jie Wang

Executive Director (Postdoc Advisor)
 Center for Sustainable Development and Global Competitiveness
 Department of Civil and Environmental Engineering
 Stanford University
 Reference email: jiewang@stanford.edu